

Visual Keyword/Result Linking: Using Interaction to Dynamically Reveal Relationships

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ABSTRACT

Keywords contain important contextual information about search results within academic digital library search interfaces. However, such information tends to be underutilized within modern search interface designs. In prior work, methods for visually linking keywords between search results have been proposed and studied. In this research, we analyze the design space and propose a new approach that aggregates the keywords over all items on the search engine results page (SERP), visually linking them back to their source search result. We have created interactive and static versions of both interfaces, and conducted a controlled laboratory study to assess the impact of the interfaces on measures of utility (efficiency, effectiveness, feature use) and perceived value (usefulness, ease of use, satisfaction, user engagement, knowledge gain, and interest gain). The findings from this research show the merit of using keywords to provide summaries of documents and search result sets, the value of making keywords interactive, and the benefit of using visualization to interactively link information within a search engine results page. The differences between providing the keywords along side each document or aggregated over the entire SERP were minimal, suggesting that it does not matter how the keywords are represented as long as they can be used to interactively reveal relationships among the search results.

CCS CONCEPTS

• **Information systems** → **Search interfaces**; • **Human-centered computing** → *Information visualization*; **User studies**; • **Interactive Information Retrieval** → *Digital Library Search Interface*.

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Keyword representation, information visualization, interactive information retrieval, exploratory search, academic digital library search, controlled laboratory study

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1 INTRODUCTION

A foundational service provided by academic digital libraries is the ability to search within their collections. Most digital library search interfaces draw upon standard practice within web search, using simple “query box and ten blue links” approaches that are tuned for simplicity and to support lookup search behaviours. While faceted navigation is often included to enable searchers to more accurately describe their information needs, little support is provided to assist searchers in assessing search results, either individually or as a set.

Among the variety of search tasks that an academic searcher may pursue, only fact verification tasks are aligned with these simple search interfaces. More advanced academic search tasks, such as knowledge enhancement, knowledge acquisition, and knowledge discovery [13, 14], are characterized as complex search tasks. What makes such tasks complex is their multi-faceted, ill-defined, and open-ended nature. That is, there may be multiple different aspects of the task that need to be pursued, the searcher may not know enough about the task to issue good queries or make good relevance decisions, and there may be no clear criteria that indicates to the searcher when they have found enough information. For such complex search tasks, the simple search interfaces designed to enable lookup search behaviour are not sufficient.

Exploratory search has been proposed as a model that is well suited to supporting complex search tasks. Within this model, a searcher starts by conducting a series of searches that are classified as *exploratory browsing*. The goal is to develop an understanding of the search task and reduce uncertainty in what they are actually seeking. When sufficient knowledge has been gained, a transition

to *focused searching* occurs wherein searchers continue to search but now with a precise goal. Searchers must use the information presented in the search interface and in the search results they access to narrow down their topic of interest, as they make this transition from exploratory browsing to focused searching.

In prior work, we proposed that keywords associated with individual search results could be used to enable exploratory browsing [28]. Clicking on keywords would create a visual link to the same keyword in other search results via colour encoding. A carefully conducted user study compared this approach to a typical baseline search interface, showing that participants were able to make effective use of the visual keyword linking, increase their perception of the usability of the interface, increase their knowledge gain, and reduce their uncertainty [12].

An open question from this prior work is whether providing the keywords alongside each search result is the best way to utilize the keyword data. In this research, we considered other alternatives and implemented an approach that aggregates the keywords for all search results on a given search engine results page (SERP) into a single list, and using interactive colour encoding to allow the searcher to visually map selected keywords back to their source search results. By disconnecting the keywords from their source and ordering them by frequency, they serve as a summary of the search results as a group, rather than a summary of individual documents.

The remainder of this paper is organized as follows. In Section 2 we review the literature that informed this research. In Section 3 we summarize the design space for keyword/document linking and the alternatives implemented for this study. In Section 4 we describe the methodology and procedures used to evaluate the search interfaces. In Section 5 we present the results of the study. In Section 6 we discuss the results and their limitations. In Section 7 we summarize the primary contributions and avenues for future work.

2 LITERATURE REVIEW

Academic searchers often undertake search tasks that are complex in nature. This complexity can be a result of the task consisting of multiple facets, being ill-defined, or its open-ended nature. A single search result is not sufficient to resolve the information need, and there is a significant element of learning that must be done [8]. In such cases, simple lookup approaches are not sufficient to help searchers complete their tasks [4]. By extension, simple search interfaces that are tuned to support lookup search approaches are also not sufficient.

While a multitude of advanced search strategies have been described in the literature [9], we focus here on exploratory search as a strategy to help searchers undertake complex search tasks. White & Roth's formulation of exploratory search [33] suggests that searchers pursuing ill-defined search tasks undertake two types of searches: exploratory browsing to learn about the task, and focused searching to find documents on a narrowly defined aspect of the task. The searching during the exploratory browsing phase focuses on investigation and discovery of new information to reduce the ill-defined nature of the task and thereby reducing the searcher's uncertainty in what they are seeking. The searching during the focused searching phase focuses on issuing narrowly

specified queries based on what has been learned, examining the results, and extracting the required information. These phases are similar to the stages in Vakkari's three-stage model of information seeking [29], where the pre-focus stage can be mapped to exploratory browsing, and the focus-formulation and post-focus stages can be mapped to focused searching [12].

While searchers may employ exploratory search strategies using simple search interfaces that are tuned for lookup search tasks, doing so places the effort of managing the exploratory search process entirely on the searcher. They may use external documents to keep track of their queries, note specific terms they have identified as being potentially relevant, and document what they have learned. They may use browser tabs as ad hoc workspaces, periodically reviewing the documents they have found and pruning out the ones that no longer match their current understanding of the topic.

Returning to the issue of how to help searchers deal with complex information needs, an open question is: Can we design search interfaces that encourage searchers to undertake the exploratory browsing needed to learn about their tasks? Since others have provided comprehensive surveys of search interface approaches [9, 23], we focus our review on work that uses keywords found within the search results.

One strategy is to aggregate the keywords in order to provide overviews of the SERP [1, 15, 16, 19, 22, 25] and using these overviews to illustrate the relationships among the search results [6, 11, 17]. One of the drawbacks of such approaches is that the keywords are disconnected from their source search results, making them difficult to use when assessing and comparing the search results themselves.

Another strategy is to keep the keywords associated with their source documents, providing a simple list of keywords for each search result [28]. While such keywords serve as good descriptors of the search results, it can be difficult to identify repeating patterns of keywords among the search result set.

A possible approach for addressing the issues of connecting aggregated keywords back to their source search results, or result-level keywords to each other, is to make the keywords interactive. Colour-encoding has been used to show keyword contributions to the ranking algorithm [7], the association between sub-queries and search results [6], and the association between query terms and search results [15]. From an information visualization perspective [31, 32], this use of similarity in visual encoding is known as the Gestalt Principle of Similarity [18], enabling searchers to perceive and interpret relationships between similarly-represented information in the interface.

3 DESIGN SPACE FOR KEYWORD/RESULT LINKING

When considering the various ways in which keywords can be used support exploratory search within an academic digital library, it is useful to consider this as a design space to be studied. A fundamental aspect of the keyword/result data is the nature of their relationship with one another. Since search results can contain multiple keywords, and a given keyword can appear in multiple results, the relationship is many-to-many. What this means is that either can be used as the unit of focus when designing a search interface.

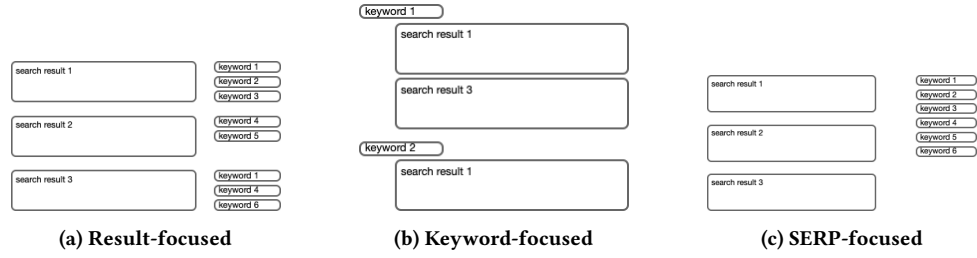


Figure 1: A summary of the candidate design space for using keywords in a search interface.

Given the long history of search interfaces providing ordered lists of search results, a logical starting point is to use the search result as the unit of focus, and list all of the keywords for each result. We call this a “result-focused” approach, and studied it in our earlier work [12, 28]. The downside of this approach is that a given keyword may be listed multiple times on a search engine results page (SERP) creating a sense of repeating information. In our earlier work, we leveraged this repetition by making the keywords interactive, allowing the searcher to visually highlight the same keyword appearing multiple times on the SERP.

Flipping things around, we could use the keyword as the unit of focus, and for each keyword list all of the search results associated with it. We call this a “keyword-focused” approach, but it has some significant drawbacks. Since a given search result may be listed multiple times on a SERP (once for each keyword with which it is associated), we have a similar repetition problem as with the “result-focused” approach. However, this repetition is even more pronounced, since the amount of information being repeated is more significant (entire search result entries) and these are the items that the searcher is evaluating in detail. Further, this repetition could generate an imbalanced SERP, given the disparity in the vertical size of a keyword compared to a search result. Given these obvious problems with the “keyword-focused” approach, we discard this line of reasoning from the design space.

If we consider that search results group keywords together, we may consider how keywords could group results together. If we aggregate all of the keywords for the results on a given SERP into a single list, these keywords represent an overview of the set of search results. Making an interactive selection of a keyword from this list could be used to highlight (visually group) all the search results that use this keyword. As the aggregation is over the full SERP, we call this a “SERP-focused” approach. The three design alternatives are illustrated in Figure 1.

Given the interactive nature of the “result-focused” and “SERP-focused” approaches for revealing the relationships within the data (between keywords in the former, and between results in the latter), we also consider static versions of these. The static version of a “result-focused” approach is very much like traditional search interfaces, providing metadata about each search result but not allowing the searcher to interact with this metadata beyond clicking on the search result to view its details. The static version of a “SERP-focused” approach also has a strong basis in traditional search interfaces, but includes a list of static keywords that serve as a summary of the current SERP.

Screenshots of these four candidate interfaces are provided in Figure 2. As the source data for these search interfaces were extracted from the IEEE Xplore Digital Library search interface, we followed their approach in terms of the metadata provided and how query terms are highlighted in the search results. Where we deviated from their approach is in how the abstracts are presented (we provide the abstract information for all search results; they hide this information in an expandable region under each search result), how the metadata is organized in a grid-based design, and how the keywords are presented (we show the keywords; they do not reveal this information on the SERP).

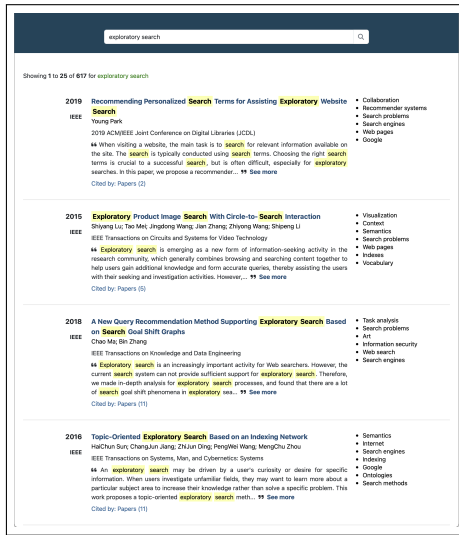
Within the source data, two types of keywords are provided: author-supplied and system-supplied. While our earlier work used author-supplied keywords, we consider the choice between these alternatives more carefully here. The author-supplied keywords give authors of source documents the autonomy to choose whichever terms they feel best describe their work; however, this means that two authors may use different terms to describe the same concept. The system-supplied keywords requires that authors of source documents choose keywords from a pre-defined vocabulary; however, this means that authors may choose keywords that only partially match the concepts in their work. Since the system-supplied keywords will provide a more consistent interface experience over time, we have chosen this approach for our current work. In the rare case where system-supplied keyword data is missing, the user-supplied keywords are used in their place. The specific ways in which the keywords are treated in the four interfaces are described in the subsections that follow.

3.1 Result-Focused, Static (RF-S)

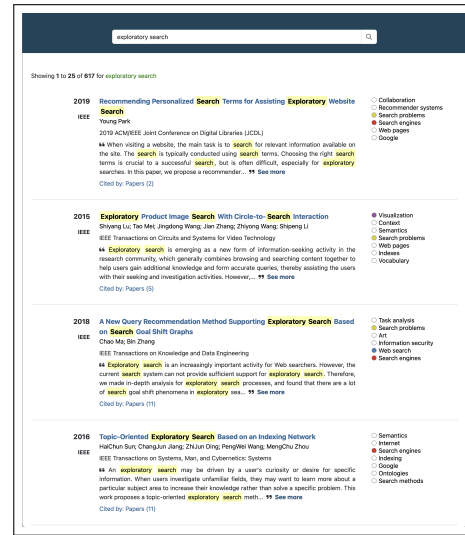
The RF-S interface (see Figure 2a) enhances the traditional list-based search results design with a set of keywords added in a vertical list beside each search result. The keywords are not interactive, serving only as metadata for each search result. As this search interface is functionally equivalent to most academic digital library search interfaces with respect to how the search results are organized, represented, and can be interacted with, we treat it as the baseline approach in this research.

3.2 Result-Focused, Interactive (RF-I)

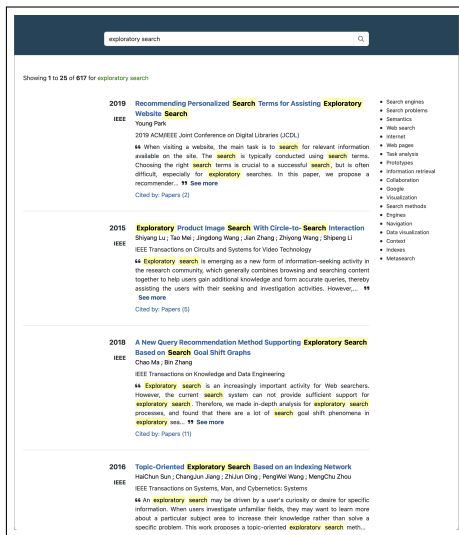
The RF-I interface (see Figure 2b) is visually very similar to the RF-S interface. There is a subtle difference in the look of the mark in front of each keyword. While this mark is bullet point in the



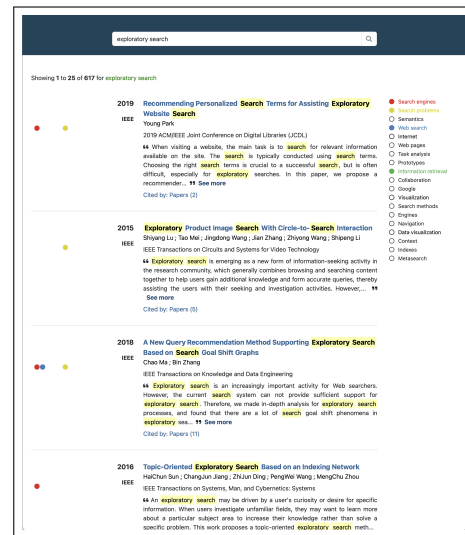
(a) Result-focused, static (RF-S)



(b) Result-focused, interactive (RF-I)



(c) SERP-focused, static (SF-S).



(d) SERP-focused, interactive (SF-I).

Figure 2: The search interfaces under investigation in this study. Note that the two interactive interfaces (b, d) show the interface after the searcher has selected four relevant keywords for their search task.

RF-S interface, it is represented as an unfilled closed disc (circle) in the RF-I interface. When searchers hover their mouse pointer over any keyword, the keyword is underlined and the pointer changes to a finger icon showing that this is an interactive element.

When a keyword is selected, the disc is filled-in with a distinct colour chosen from a set of ten visually unique colours (we use only ten colours due to the limits of the human visual perception system with respect to qualitative differentiation [31]). In addition to marking the selected keyword in this way, the same keyword in all other search results on the SERP are marked with the same colour.

This serves as a visually linking approach, allowing the searcher to easily observe this same keyword across all other search results (e.g., the keyword “Search engines” can be quickly identified in the first, third, and fourth search results in Figure 2b). It provides a mechanism that supports the searcher in learning about what is present on the SERP and how common the keywords of interest are among the search results.

In addition to supporting the searcher to find documents that use the selected keywords (visual scanning of the keywords for filled-in discs), the searcher can also easily identify the search results for

which no selected keywords are present (visual scanning for non-filled-in discs). This serves as a mechanism to support the discovery of search results and their associated keywords that are different than the keywords that are currently selected [12].

3.3 SERP-focused, Static (SF-S)

The SF-S interface (see Figure 2c) uses the traditional list-based search results design as the basis, but with the addition of a single list of keywords aggregated from the full set of search results on the page. These are ordered by the frequency of the keyword (the most frequent first). The list of keywords is not interactive, but it does serve as an overview or summary of what is present on the SERP. Note that the width of the page used to display the search results is slightly narrower than the result-focused interfaces (RF-S and RF-I) in order to reserve space for the keyword-result linking in the interactive version of this approach (SF-I). That same space is reserved here for consistency between these interfaces.

3.4 SERP-focused, Interactive (SF-I)

The SF-I interface (see Figure 2d) is visually similar to the SF-S interface. The visual differences are the same as described earlier for the RF-I interface: an unfilled closed disc and visual cues for interactivity are provided, and when a keyword is selected the disc is filled-in with a qualitatively unique colour chosen from a set of ten such colours.

What is different here is how this keyword selection is used to link it back to its source search result(s). This is where the space reserved to the left of the search results is used. Ten hidden columns are reserved in this space, representing the maximum of ten keywords that can be selected. Whenever a keyword is selected, it is assigned a colour and column in this space, and for each search result that includes this keyword, a corresponding filled-in disc is shown. This can be observed in Figure 2d, where the “Search engines” keyword is selected and assigned the colour red, and red discs are provided beside the first, third, and fourth search result.

The alignment of the discs to show the relationship to the selected keywords is important for supporting visual scanning for search results that use a specific keyword. The relationship is redundantly encoded using both colour and spatial alignment of the disc. This enables the searcher to scan in a straight line along the column associated with the selected keyword, making observations of whether the associated result uses the keywords based only on the existence of the colour-encoded disc. If we had packed the discs together for each search result, these would have been treated visually as a group and would have required the searcher to spend more cognitive effort visually decoding them.

The SF-I interface provides the searcher with a mechanism for learning about the relationship between the keywords and the search results through interactive selection. Further, like the RF-I interface, after selecting a set of relevant keywords, searchers can visually scan the SERP to identify search results that are not using any of these (the absence of discs beside the search result) in order to discover search results that may contain unknown information.

4 EVALUATION METHODOLOGY

4.1 Research Questions

The evaluation of the four approaches to supporting the use of keywords within an academic digital library search interface was driven by the following two research questions:

RQ1: How does the method of representing and interacting with the keywords impact the utility of the search interface?

RQ2: How does the method of representing and interacting with the keywords impact the perceived value of the search interface?

4.2 Study Design

A controlled laboratory study was designed in which we manipulated one independent variable and measured its impact on a set of dependent variables. The single independent variable in this study is the method for representing/interacting with the keywords within the search interface. It has four levels, corresponding to the four search interfaces described in the previous section: (1) result-focused, static (RF-S); (2) result-focused, interactive (RF-I); (3) SERP-focused, static (SF-S); and (4) SERP-focused, interactive (SF-I). This independent variable was treated using a within-subjects design. In order to mitigate potential learning effects created by the participants using each of the search interfaces, a balanced 4x4 Latin square was used to vary the order of exposure. Since the four interface alternatives all use a typical search interface as their basis and simple methods for representing the keywords, we expect the learning effects to be minimal. As a result, the within-subjects design allows us to collect four datasets for each participant recruited into the study (rather than one data set if we had used the alternative approach of a between-subjects design).

The dependent variables in this study are utility and perceived value (corresponding to the two research questions). We operationalized utility with three measures: (1) efficiency; (2) effectiveness; (3) feature use. Efficiency is measured as time to task completion. Effectiveness is measured as the precision of the documents found in the search process. Feature use is measured by logging which features are used throughout the search process using LogUI [24].

We operationalized perceived value with six measures: (1) perceived usefulness; (2) perceived ease of use; (3) perceived satisfaction; (4) perceived user engagement; (5) perceived knowledge gain; and (6) perceived interest gain. Perceived usefulness and ease of use were measured using four survey items each, adapted from the TAM2 (Technology Acceptance Model) instrument [30]. Perceived satisfaction was measured using four survey items that are similar in nature to the TAM2 questions. Perceived user engagement was measured using the UES-SF (User Engagement Scale - Short Form) instrument [26]. All of these were administered after the participant used each interface to complete a search task. Perceived knowledge gain and interest gain were measured using single-statement survey questions administered before and after the search tasks. All survey data was collected on 5-point Likert scales. The survey instruments used in this study are provided as part of the supplemental material provided within the ACM Digital Library.

4.3 Simulated Search Tasks

Using a within-subjects design with four levels for the independent variable implies the need for four similarly complex search tasks in order to provide experimental control over this aspect of the study. Following the guidance of Kules & Capra [20] regarding the creation of search tasks for exploratory search systems, the search tasks were designed to have the following features: (1) they indicate uncertainty that requires discovery of new information; (2) they are in a domain that is interesting but unfamiliar for the searchers; and (3) they provide sufficient descriptions to enable the searchers to relate to the search context.

As we were recruiting participants from the undergraduate computer science student population at our University, we formulated the academic search tasks in the context of taking specific advanced courses in computer science/software engineering and needing to find academic literature to support the writing of a research report. In order to control the depth at which participants would undertake their exploratory search activities, we instructed the participants to conduct the exploratory browsing aspect of exploratory search, with the goal of learning enough about the topic to proceed with focused searching at a later date. While this limited the ability for participants to search using their natural behaviour, it provided the consistency needed for a controlled laboratory study.

The search tasks were presented in the format of simulated work task situations [5]. They all follow a common structural format similar to our earlier study [12]. The specific tasks are provided in Figure 3. Each was verified to have a sufficiently broad coverage in the underlying academic search engine used by the search interfaces (IEEE Xplore). A caching mechanism ensured that all participants received the same set of search results for the same query.

The same training task was used when introducing each new search interface. It followed a similar pattern as the assigned search tasks, with a focus on a machine learning course and the topic of “machine learning approaches for detecting brain disease”. During the training tasks, we described a specific process for conducting the exploratory browsing style of searching, using the tab feature of the browser as an ad hoc workspace (the search interfaces were all programmed to open documents in new tabs). The specific instructions were:

- (1) Issue a query about the topic to start.
- (2) Evaluate the search results to find and open in new tabs some papers that you think are relevant.
- (3) Review these search results you have opened in new tabs, close those that you no longer think are relevant.
- (4) Submit a new query based on what you have learned.
- (5) Repeat this process of evaluating and reviewing search results, reconsidering what you have opened so far, and re-querying to find new information.
- (6) You can repeat this loop as many times as it takes for you to feel confident that you have learned and opened enough papers in new tabs to read them in detail later.

The search sessions were open-ended, with no time limit given. In order to control for potential learning effects associated with the tasks, a second balanced 4x4 Latin square was used to vary the order of the tasks assigned to the participants.

Task 1: Suppose you are enrolled in a course on **Information Retrieval**. For this course, you need to write a research report on **image search engines**. Indexing images and matching them to queries can be done in a variety of ways. Your goal is to explore the research literature to learn more about this topic and to choose a specific focus for your paper. You do not need to perform a deep or comprehensive search; instead your goal is to get to the point where you have a few articles that you can use as a start for a more focused search that you will perform later.

Task 2: Suppose you are enrolled in a course on **Data Mining**. For this course, you need to write a research report on **pattern recognition in big data**. Various advanced analytics techniques can be used to identify patterns and relationships in large or complex data sets. Your goal is to explore the research literature to learn more about this topic and to choose a specific focus for your paper. You do not need to perform a deep or comprehensive search; instead your goal is to get to the point where you have a few articles that you can use as a start for a more focused search that you will perform later.

Task 3: Suppose you are enrolled in a course on **Artificial Intelligence**. For this course, you need to write a research report on **sentiment analysis in text mining**. There are several methods to identify and categorize opinions expressed in a text, particularly when determining whether the writer’s attitude toward a particular topic is positive, negative, or neutral. Your goal is to explore the research literature to learn more about this topic and to choose a specific focus for your paper. You do not need to perform a deep or comprehensive search; instead your goal is to get to the point where you have a few articles that you can use as a start for a more focused search that you will perform later.

Task 4: Suppose you are enrolled in a course on **Computer Networks**. For this course, you need to write a research report on **mobility management in wireless networks**. Specific techniques should be used to track where subscribers are and whether to allow calls, SMS, and other mobile phone services to be delivered to them. Your goal is to explore the research literature to learn more about this topic and to choose a specific focus for your paper. You do not need to perform a deep or comprehensive search; instead your goal is to get to the point where you have a few articles that you can use as a start for a more focused search that you will perform later.

Figure 3: Assigned academic digital library search tasks.

4.4 Procedures

The study procedures were built into a Qualtrics survey with links to the search interfaces. The two 4x4 Latin squares for manipulating the order of exposure to the search interfaces and the assignment of search tasks were programmed into the survey.

To initiate the study session, we provided a consent form that described the purpose of the study, its procedures, the risks and benefits, how the data would be protected, and the rights of the participant to withdraw at any time. Implied consent was collected as the participant proceeded to the next page of the survey. A pre-study questionnaire was administered that collected demographic data, level and major of study, English language proficiency, and academic search experience. This data was used to verify that the participants formed a reasonably coherent population of students in computer science/software engineering.

For a selected search interface, a link to a training video was provided. The participants were then provided with a written description of the same training task, and were given the opportunity to use the search interface to replicate the search themselves.

The assigned search task was then presented to the participant. A pre-task questionnaire was administered to collect the participants’ perceived prior knowledge and interest in the task. The participants were asked to use the assigned search interface to undertake the assigned search task. The time to task completion, documents

viewed, and search interface features used were logged. A post-task questionnaire was administered to collect the same data as in the pre-task questionnaire, along with questions related to perceived usefulness, ease of use, satisfaction, and user engagement.

The participants were instructed to take a short stretch break after each task, which mitigated the effects of fatigue and served as a cognitive reset. This process was repeated using the other three search interfaces and performing the other three search tasks. The entire study took approximately one hour to complete. As compensation for their time, participants were given an option to be entered into a draw for a set of high-end wireless headphones or to be given research participation credit in a course they were taking. The study procedures and documents were reviewed and approved by the Research Ethics Board at our University.

The study sessions were administered in two modalities: asynchronous and synchronous. Initially, we wished to collect all data asynchronously. However, we discovered that a large number of participants were abandoning the study prior to completion. It appeared that some were accidentally closing the browser window that contained the online survey controlling the study as they transitioned from one search interface/task to the next, and were having a difficult time resuming the study. This did not affect the data collected for complete study sessions. To address this issue, we started running the data collection sessions synchronously, with a member of our research team being present in a video call with the participants to monitor their activities and guide them between the search tasks. No changes were made to the data collection procedures between the asynchronous and synchronous modalities, and no differences were found between their respective datasets.

4.5 Participants

We recruited participants for this study from among the students at our University who were majoring in computer science/software engineering programs. In order to ensure that the participants had sufficient prior knowledge to make the simulated work task scenarios realistic, we required that each participant be in at least the third year of their undergraduate program or the first year of a graduate program. This participant group represents a sub-population of academic searchers who are technically adept and topically knowledgeable, but with a moderate level of experience with academic digital library searching.

Although the two 4x4 Latin squares used to vary the order of exposure to the search interfaces and tasks implies that participants should be recruited in multiples of 16 to fully balance the conditions, we stopped our data collection at 24 participants. Intermediate analysis revealed a balance in the distribution of prior knowledge and interest between the search interface conditions and statistically significant results between the search interface conditions on multiple measures. As a result, we considered the varying of the search interface order as the primary 4x4 Latin square needed to manage learning effects, verifying that this provided a full balance of the order of interface exposure across the 24 participants.

Among these participants, 10 completed the study asynchronously and the remainder completed it synchronously. Given a wide range of choices for identifying their gender identity, 6 participants identified as women and 18 participants identified as men. The average

age of the participants was 22.5 years old (standard deviation: 4.1). Regarding the year of study, 11 participants were in their third year, 9 participants were in their fourth year, and 4 participants were in their first year of graduate studies. All participants reported moderate or better English language proficiency. In terms of their self-reported skills in searching within academic digital libraries, the responses ranged from somewhat bad to extremely good, with a median at the somewhat good level. There was also a range of responses regarding the frequency of searching within an academic digital library, with the median at the once a week level. We consider this a representative sample of the student population who might need to conduct the complex search tasks used in this study.

4.6 Data Analysis

The data collected for the utility measures (RQ1) are all numerical data: efficiency measured via time to task completion; effectiveness measured via precision of the saved documents; and feature use measured by counting how many times each feature type was used (input, information, control, and personalization [34]). Statistical significance between the experimental conditions was analyzed using two-tail ANOVA tests with a p-value of 0.05.

The treatment of precision requires special attention. All of the documents saved in browser tabs by the participants at the end of the search task (indicating an interest in preserving them in the tab-based workspace) were assessed for relevance to the assigned search tasks by the first two authors independently. A 3-point relevance scale was used: 0 = not relevant; 1 = partially relevant; 2 = fully relevant. Inter-rater reliability was calculated using Krippendorff's Alpha. A strong level of agreement was found ($\alpha = 0.86$). Resolutions of disagreements were achieved through discussions. This data was converted to binary relevance assessments by merging the partially relevant and fully relevant scores. While this optimistic view of relevance is counter to some recommendations [27], it is appropriate for this particular study since participants were instructed to find search results they thought might be relevant (rather than those they were sure were fully relevant). With these binary relevance scores, a precision score was calculated in the typical way for each task a given participant undertook.

The data collected for the perceived value measures (RQ2) are all Likert scale scores. For each of the perceived usefulness, ease of use, and satisfaction concepts, four questions were asked. For the perceived user engagement concept, 12 questions were asked. While such raw data is ordinal in nature, a common practice is to aggregate the scores for each concept (mean value), transforming the data into numerical values for which ANOVA tests can be run [21]. The data was verified to be sufficiently close to being normally distributed to warrant the use of the ANOVA test. Two-tailed tests were used, with a p-value of 0.05.

For the perceived knowledge gain and interest gain measures, one question was asked for each, both before the search task was started and after it was completed. Differences in the mean values were computed and statistical significance was tested using Mann-Whitney U tests. Statistical significance between the experimental conditions on the calculated perceived knowledge and interest gain values were tested using Kruskal-Wallis tests. Both used a p-value of 0.05.

5 RESULTS

5.1 RQ1: Utility Measures

The data and statistical analysis for the measures of efficiency (time), effectiveness (precision), and feature use (counts) are provided in Table 1. Statistically significant differences were found between the experimental conditions on two of these measures (time to task completion, feature use: information). For each of these, post-hoc analysis (Tukey HSD) was used to identify between which conditions the differences were significant.

The measure of the time to task completion showed a pattern in which using the interactive interfaces (RF-I and SF-I) resulted in spending more time searching. This is as one would expect, given that using the interactive features to highlight keywords (RF-I) or mark the source search result of a selected keyword (SF-I) takes time to perform and consider the results. While statistical significance was found when comparisons were made over the four conditions, Tukey HSD did not reveal statistically significant differences between the interactive and static conditions (RF-S v. RF-I; SF-S v. SF-I), nor between the result-focused and SERP-focused conditions (RF-S v. SF-S; RF-I v. SF-I). We conclude that the pattern of increased time spent using the interactive interfaces is not significant, and that there is no difference in the time it takes to use the keywords when they are provided alongside each search result versus using the keywords when they are aggregated over the SERP and provided in a single list. That is, there was no penalty incurred by interacting with the keywords.

The precision of the tabbed documents was similar across all four experimental conditions, with slightly higher precision achieved for the interactive interfaces. The differences were not found to be statistically significant. Since all search interfaces used the same underlying academic digital library, and the participants issued similar queries derived from the assigned search tasks, the search results provided across the experimental conditions were very similar. While it is encouraging to see that the participants made better choices among documents to save in the tabs when they were able to interact with the keywords to reveal related search results, the effect was not sufficiently significant to allow strong conclusions to be drawn with respect to the utility of the interactive interfaces.

Regarding the feature use, there were small differences in the input (issuing queries) and control (moving to the next page of search results) features between the experimental conditions. However, for the information features (viewing the full abstract, clicking through to the document), these features were used more on the interactive interfaces (RF-I and SF-I) than on the static interfaces (RF-S and SF-S). Statistical significance was found across all four conditions, but post-hoc analysis using Tukey HSD did not reveal significance when comparing the interactive and static conditions nor the result-focused and SERP-focused conditions. So, while the interactive conditions lead the users to interact with the search results to a higher degree, the effect was not significant.

As there were no personalization features in the two static conditions (RF-S and SF-S), we focus on how often these features were used for the interactive conditions (selecting and un-selecting keywords in the RF-I and SF-I interfaces). While it is clear that participants were engaged in using these features when they were present in the interfaces, there was some variability within the

data (relatively high standard deviation). No statistically significant differences were found, leading us to conclude that the ability to interact with the keywords was supported equally well regardless of whether the keywords were provided with each search result (RF-I) or provided in a single list for the full SERP (SF-I).

5.2 RQ2: Perceived Value Measures

The data and statistical analysis for the measures of perceived usefulness, ease of use, satisfaction, and user engagement are provided in Table 2. Statistically significant differences were found between the experimental conditions on all four measures. Post-hoc analysis (Tukey HSD) was used to identify between which conditions the differences were significant.

For perceived usefulness, participants reported the interactive versions of the interfaces to be more useful (RF-I > RF-S, $p < 0.01$; SF-I > SF-S, $p < 0.05$), but did not report statistically significant differences between the result-focused and SERP-focused interfaces. Similar patterns were found for the other three measures (perceived ease of use: RF-I > RF-S, $p < 0.0001$; SF-I > SF-S, $p < 0.0001$) (perceived satisfaction: RF-I > RF-S, $p < 0.005$; SF-I > SF-S, $p < 0.01$) (perceived user engagement: RF-I > RF-S, $p < 0.05$; SF-I > SF-S, $p < 0.05$). In addition, participants reported that the SERP-focused static interface to be easier to use than the result-focused static interface (SF-S > RF-S, $p < 0.05$). Without being able to interact with the keywords, the participants may have found the single set of keywords presented for all the search results (see Figure 2c) to provided a more structured overview of the search results than the individual sets of keywords beside each search result (see Figure 2a).

While there was a general pattern of the participants reporting higher scores for the SERP-focused conditions over the result-focused conditions (SF-S > RF-S; SF-I > RF-I), the differences between these were not statistically significant. So, while we conclude that the interactive interfaces were superior to the static interfaces, we cannot draw any strong conclusions from this data regarding whether it would be better to represent the keywords per-search result or per-SERP. What we can say is that participants found the visual method for interactively linking the data together (keyword to keyword in RF-I; keyword to search result in SF-I) to be much more valuable than just representing the keywords statically in the interface, and that as long as this was possible it did not matter to them in what format the keywords were provided.

For the perceived knowledge and interest measures, we start by analyzing the differences when comparing the prior- and post-task data, representing perceived knowledge gain and interest gain. The data and statistical analysis for these measures are provided in Table 3. Participants reported statistically significant increases in perceived knowledge after using three of the four interfaces (all but the SF-S interface). Focusing on the two interactive interfaces (RF-I and SF-I), after completing the searches with these interfaces, participants had similarly higher impression of their knowledge of the topic. Where things differed was in the prior knowledge, which was affected by the randomness of how much knowledge participants possessed about the assigned tasks.

Of note, the SERP-focused static interface (SF-S) did not result in participants perceiving increased knowledge gain. Because the keywords were disconnected from the search results and there

Measure	RF-S	RF-I	SF-S	SF-I	ANOVA
time to task completion	8:17 (1:49)	9:40 (1:43)	7:55 (2:30)	9:08 (2:38)	$F(3, 92) = 3.11, p < 0.05$
precision of tabbed documents	0.83 (0.30)	0.96 (0.11)	0.84 (0.25)	0.87 (0.20)	$F(3, 92) = 1.52, p = 0.215$
feature use: input	3.58 (1.14)	3.38 (1.58)	3.46 (1.74)	2.75 (0.94)	$F(3, 92) = 1.71, p = 0.171$
feature use: information	4.17 (1.81)	5.08 (1.44)	3.88 (1.39)	5.04 (2.16)	$F(3, 92) = 3.02, p < 0.05$
feature use: control	1.33 (1.81)	1.58 (1.86)	1.17 (2.01)	1.21 (1.91)	$F(3, 92) = 0.233, p = 0.873$
feature use: personalization	-	8.54 (7.11)	-	8.25 (5.39)	$F(1, 46) = 0.026, p = 0.874$

Table 1: The mean (standard deviation) data for time to task completion (minutes:seconds), precision (ratio), and the number of times the four types of features were used. Statistically significant differences were identified using ANOVA.

Measure	RF-S	RF-I	SF-S	SF-I	ANOVA
usefulness	3.67 (1.25)	4.50 (0.61)*	4.10 (1.06)	4.82 (0.35)*	$F(3, 92) = 7.54, p < 0.0005$
ease of use	3.54 (1.09)	4.45 (0.55)*	4.18 (0.89)**	4.64 (0.55)*	$F(3, 92) = 8.72, p < 0.0001$
satisfaction	3.66 (1.05)	4.47 (0.54)*	3.79 (0.93)	4.54 (0.52)*	$F(3, 92) = 7.85, p < 0.0005$
user engagement	3.48 (0.97)	4.10 (0.57)*	3.53 (0.83)	4.22 (0.58)*	$F(3, 92) = 6.06, p < 0.001$

Table 2: The mean (standard deviation) scores for usefulness, ease of use, satisfaction, and user engagement. The data was measured on 5-point Likert scales. Statistically significant differences were identified using ANOVA, with * indicating where significant differences were found between the static and interactive versions of a given approach for representing the keywords, and ** indicating where significant differences were found between the approaches.

Interface	Perceived Knowledge Gain				Perceived Interest Gain			
	Prior	Post	Δ	Mann-Whitney U	Prior	Post	Δ	Mann-Whitney U
RF-S	2.96 (1.3)	3.75 (1.07)	0.79	$V = 0.00, p < 0.001$	3.62 (1.13)	3.88 (1.08)	0.26	$V = 30.0, p = 0.12$
RF-I	2.96 (1.2)	3.79 (0.98)	0.83	$V = 0.00, p < 0.005$	3.58 (1.28)	3.88 (1.19)	0.3	$V = 20.0, p = 0.13$
SF-S	3.12 (1.15)	3.50 (0.88)	0.38	$V = 20.5, p = 0.14$	3.58 (1.25)	3.75 (1.07)	0.17	$V = 40.5, p = 0.75$
SF-I	3.17 (1.24)	3.79 (0.93)	0.62	$V = 13.0, p < 0.005$	3.71 (1.16)	4.08 (0.88)	0.37	$V = 9.0, p = 0.052$

Table 3: The mean (standard deviation) scores for knowledge and interest, collected pre- and post-task. The data was measured on 5-point Likert scales. Statistically significant differences were identified using Mann-Whitney U.

was no interactive way of linking these to their source results, this information served only as a summary of the search results set rather than a tool that can be used to gain knowledge. On the other hand, for the result-focused static interface (RF-S) interface, the keywords remained associated with their source search result, allowing the searchers to observe and learn from these keywords when finding relevant search results.

While there were small increases in interest after using each interface, none were statistically significant. This is not unexpected, given that the tasks were assigned to the participants rather than chosen by them, and a relatively short time was given for the participants to immerse themselves in the search process.

In order to assess whether the experimental conditions had an impact on the knowledge gain and interest gain measures, statistical analysis was conducted on the delta values between the conditions. Kruskal-Wallis tests revealed no statistical significance on either measure (knowledge gain: $H(3) = 3.02, p = 0.389$; interest gain: $H(3) = 0.430, p = 0.934$). So while it was promising to see that the participants gained knowledge as a result of performing the assigned search tasks, the differences between the conditions were not sufficient to make a claim of one interface being superior to the others. As with the analysis of the precision measure, this may be because the participants were exposed to the same underlying

academic search engine, and therefore were being presented with the same information upon which to base their knowledge and interest gains.

6 DISCUSSION

In consideration of RQ1 (impact on utility), these findings show that when using the same underlying search data and typical “query box and ten blue links” search interface design, small changes to the interface such as adding keywords and making them interactive had impacts on the utility of the search interface but not to the level of statistical significance. The utility measures for the static versions of the search interfaces were rather similar, showing that participants used these in a similar manner (observing the keywords but not making much use of them). However, for the interactive versions, we measured significant use of the keyword selection features, increased access of individual search results, and longer search times. We view these all as positive outcomes for exploratory search tasks. Whether the keywords were presented in a result-focused or SERP-focused manner made little difference on the measures of utility.

In consideration of RQ2 (impact on perceived value), adding keywords and making them interactive impacted the participants’

subjective impressions of the interface. While the differences between the result-focused and SERP-focused interfaces were minimal, the interactive versions of the interfaces were reported to have significantly higher scores than the static version across all subjective impression measures (usefulness, ease of use, satisfaction, user engagement). The ability to interact with the keywords was clearly valued by the participants. Perceived knowledge gain and interest gain was found across all interface alternatives, but without significant differences between them. Perhaps the short time frames between the pre- and post-task questionnaires limited the ability for participants to perceive any changes in their knowledge or interest. Like RQ1, how the keywords were presented had much less of an impact on the perceived value than making them interactive.

The positive results when searchers are provided with a means to interact with the keywords highlights two key outcomes of this research. Firstly, it shows that giving users the ability to interactively personalize their views of the SERP are appreciated and valued. Secondly, it shows that the mechanisms for visually linking the keywords to one another (result-focused approach) and visually linking the keywords back to their source search results (SERP-focused approach) are effective communication mechanisms that support exploratory search.

Prior work on using keywords to support exploratory search activities found similar positive results [1, 6, 12, 15, 16, 25]. What is unique about the work reported here is the isolation of interactivity as a critical element leading to the positive results. This is consistent with the literature on the value of adding interactivity to the SERP [3, 10, 34] and on using information visualization techniques to enhance interactive information retrieval systems [11]. While much of the research on search interface innovations conflates the surfacing of information and the interaction with this information (including the work of the authors of this paper), this is the first study to our knowledge that compares the same search interface alternatives with static and interactive versions. While it may seem obvious that interactivity will add value, it does come at a potential usability and efficiency cost. Designing for simple and intuitive interactivity is paramount.

What we have learned in this research is that keywords contain useful and valuable information, but in order to extend that utility and value to the searcher we need to make them interactive. When such interactivity enables searchers to reveal relationships between search result through visual keyword/result linking, searchers will make use of such features resulting in increases in subjective impressions of the search interface and process. A comprehensive consideration of the design space for structuring the keyword data in relation to their search result sources and the evaluation of the candidate designs shows that there is little difference for the searchers - as long as the keywords can be used, rather than just viewed. There is a subtle difference between the result-focused and SERP-focused design that mirrors the differences between exhaustive and economical searching [2]. Keeping the keywords spatially connected to their source search results (result-focused) enables exhaustive searching, whereas providing the keywords as a separate list (SERP-focused) enables economical searching. As exhaustive versus economical searching is a personal preference [2], how this

impacts searcher performance on these search interface alternatives warrants further study.

As with all studies, this one is not without its limitations. The tasks were focused specifically on exploratory browsing style of searching, which limits the external validity and our ability to generalize the findings to other search contexts and strategies. We can expect that for simple fact-finding tasks and look-up strategies, the keywords will provide less value than search activities that require learning and investigation. While providing assigned search tasks and prescribed search processes enabled the control needed in a study of this type and contributed to internal validity, it limits the ecological validity of the findings. We mitigated this limitation by making realistic academic search tasks that were aligned with the academic study of the target participants.

7 CONCLUSION

This research makes a step in the direction of addressing the fact that document keywords are understudied as an interactive mechanism to enable searchers to take an active role in their search process. While these results are consistent with earlier work [12], by separating the visual linked keyword approach from other search interface features (e.g. a workspace), we can focus on isolating the value provided for the activity of evaluating the search results sets. Further, by considering the design space for keyword/result linking more fully, we can also determine whether there are other ways of surfacing the keyword information to the searcher to enhance their search experience. Although participants used and experienced the two ways the keywords were represented similarly, subtle differences were found with the static versions that suggest avenues for further investigation.

More importantly, we found that the key benefit of providing keyword information to searchers is to make them interactive so that they can dynamically reveal relationships among the search results. Between the two interactive versions, whether the keywords were presented in a result-focused way or a SERP-focused way did not matter. What mattered was that the interactivity could be used to highlight similarities and differences among the search results (either via keyword-keyword mapping or keyword-result mapping). This highlights the value of the interactive visual keyword/result linking approaches used in these interfaces.

There remains a broad body of research yet to be done in studying the value of using keywords to support interactive information retrieval. The trade-offs between author-supplied and system-supplied keywords has yet to be fully explored in the context of supporting learning during the search process. Drawing upon the full range of the Gestalt Principles [18], other ways of visually linking keywords/results could be studied. There is also space to combine the result-focused and the SERP-focused approaches presented in this paper into a unified approach. Whether these findings generalize to search tasks of varying complexities, and to search contexts beyond academic digital libraries remains to be studied.

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